

ASIA ECONOMICS ANALYST

RMB Internationalization Revisited: Beyond Trade Settlement

China in the Long Run

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- RMB international use has improved but still trails China's global footprint, implying solid potential to grow in the future. RMB invoicing in goods imports among sample countries has roughly doubled since 2020 but remained only around 1.5% in 2023; the RMB's share in global payments was around 3–4% over the past two years; and the RMB accounted for around 2% of global official reserves and 0.9% of international debt securities by currency of denomination in 2025. The clearest progress is in China-linked trade settlement: RMB settlement in China's goods trade rose from 13% in 2019 to 30% in 2025. With policymakers placing more emphasis on RMB internationalization, we provide a progress check and discuss what it would take for RMB to move beyond China-linked trade settlement.
- Cross-border RMB use is becoming more investment-driven. Total cross-border RMB transactions rose sharply from RMB9tn in 2017 to RMB64tn in 2024, and capital- and financial-account transactions now account for around 75% of the total. Bond investment is the largest single use case, accounting for 46% of cross-border RMB payments in 2024, compared with 19% for goods trade and 13% for direct investment. This shift means the next phase will require more than settlement channels: foreign participants need RMB liquidity that is easier to fund, tools that are easier to hedge with, and RMB assets that are more attractive to hold.
- The next phase is likely to be offshore-led and Hong Kong-centered. Rather than relying first on broad capital-account liberalization, RMB internationalization is likely to proceed through gradual onshore opening and deeper offshore RMB markets. Offshore RMB liquidity has become cheaper and more stable, helped by RMB appreciation, policy support and official infrastructure such as PBOC swap lines and offshore clearing banks. Continued RMB appreciation should help keep onshore-offshore funding gaps narrow and stable.
- Risk-management tools and RMB asset supply remain key bottlenecks. FX hedging is relatively more developed: the RMB's share of global OTC FX derivatives turnover rose from 1.0% in 2010 to 8.1% in 2025. Rates hedging is

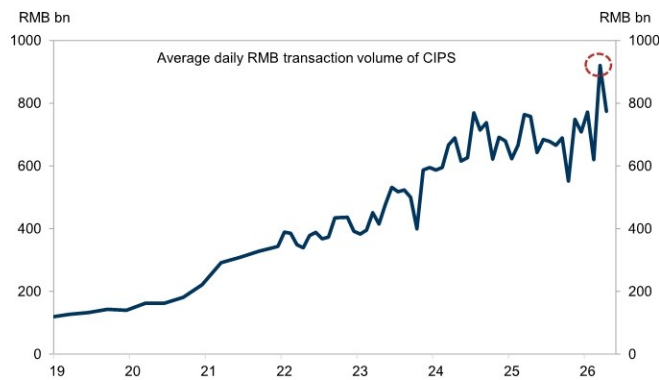
less complete: RMB's share of global OTC interest-rate derivatives turnover was only 0.8% in 2025. IRS liquidity is concentrated in the front end and 5y. Offshore investors only have limited access to CGB futures. These factors constrain offshore investors' capacity to hedge longer-dated duration risks.

- On the asset side, broader RMB asset use would require wider onshore market access, such as Connect programs covering onshore REITs and commodity products, and deeper offshore RMB markets, including a more liquid offshore RMB bond yield curve, a larger dim sum bond market, and a potential Hong Kong commodity ecosystem with more RMB-denominated products.

RMB Internationalization Revisited: Beyond Trade Settlement

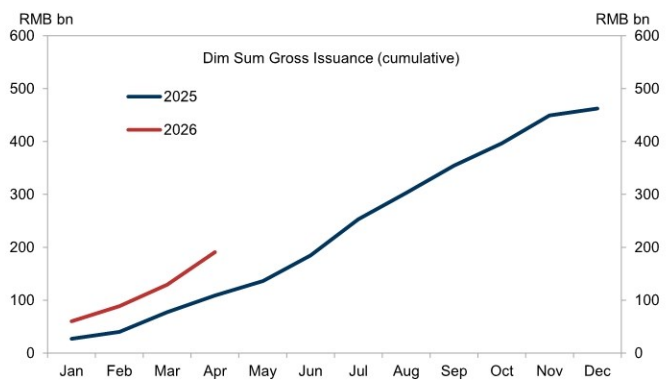
Recent developments have brought RMB internationalization back into focus. Policy communication has become more supportive, with the 15th Five-Year Plan calling for further RMB internationalization, deeper capital account opening, and a more resilient and self-reliant RMB cross-border payment system. Average daily RMB transaction value through the Cross-Border Interbank Payment System (CIPS) reached a record high in March ([Exhibit 1](#)). Against this backdrop, discussions around the “petroyuan” have resurfaced amid the Middle East conflict. Meanwhile, market conditions have turned more favorable for RMB assets: the RMB has appreciated around 3% year-to-date, more than offsetting the roughly 2.5% annualized cost of holding RMB assets implied by US-China interest rate differentials. Separately, dim sum bond (offshore RMB-denominated bonds) issuance has rebounded strongly this year ([Exhibit 2](#)).

Exhibit 1: Average daily RMB transaction volume of CIPS spiked in March



Source: CIPS, Data compiled by Goldman Sachs Global Investment Research

Exhibit 2: Dim sum bond issuance nearly doubled from a year ago in Jan-Apr 2026



Source: Bloomberg, Goldman Sachs Global Investment Research

In this context, clients have increasingly asked what these developments mean for RMB internationalization and what could drive the next phase. In this Asia Economics Analyst, we assess the progress in RMB internationalization across the three functions of an international currency – unit of account, medium of exchange and store of value – and argue that the progress remains uneven and largely concentrated in China-linked trade settlement. We discuss the market foundations needed for broader foreign participation: more stable offshore RMB liquidity, better RMB risk management tools and a broader RMB asset pool.

Where RMB internationalization stands?

To assess where RMB internationalization stands, we look at both the RMB's role across the three functions of an international currency and the composition of cross-border RMB transactions.

1. Uneven progress led by China-related settlement

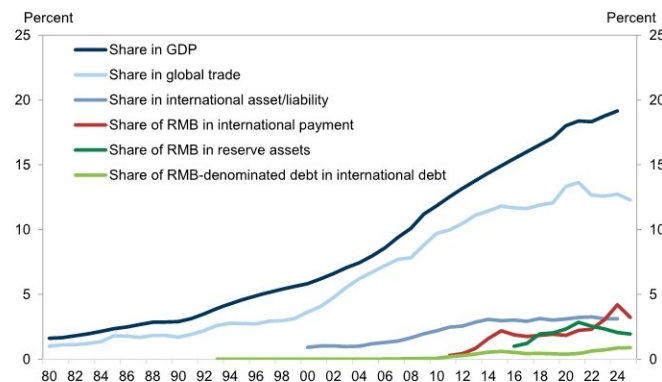
We start with the standard framework for an international currency: unit of account, medium of exchange and store of value (see our previous notes on the progress of RMB internationalization in [2023](#) and [2025](#)). In practice, this means looking at the extent to which the RMB is used in invoicing and pricing, settlement and payments, and reserve or asset holdings. We measure these roles using the RMB's share in trade invoicing,

international payments, official reserve assets and international debt market.

RMB international use has improved across functions but still has significant room to grow, especially compared to China's global share of GDP and trade ([Exhibit 3](#)). On unit of account, invoicing data are limited, but Boz et. al. (2025) [suggest](#) that the average RMB invoicing share in goods imports among sample countries remained only around 1.5% in 2023. On medium of exchange, the RMB's share in global payments has risen, but has remained at around 3-4% over the past two years. Both remain low compared with China's 19% share of global GDP in 2024 and 12% share of global trade in 2025. On store of value, the RMB accounted for around 2% of global official reserves and 0.9% of international debt by currency of denomination in 2025. Taken together, these indicators point to continued progress, but also significant room for further RMB internationalization.

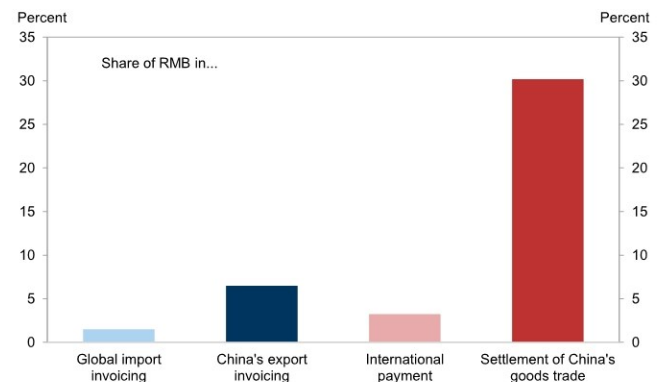
Zooming in on the relatively more advanced medium-of-exchange and unit-of-account roles, progress is still mostly China-linked RMB transactions ([Exhibit 4](#)). The share of RMB settlement in China's goods trade has risen from 13% in 2019 to 30% in 2025, while the share of RMB invoicing in goods trade with China remains below 10%. Both are much higher than the RMB's shares in global trade invoicing and international payments. Further progress would require continued growth in RMB invoicing and settlement in China-linked trade, as well as broader RMB use in third-party invoicing and settlement.

Exhibit 3: RMB international use has improved, but still trails China's global GDP and trade shares



Source: BIS, Lane & Milesi-Ferretti (2018), Haver Analytics, Wind, Goldman Sachs Global Investment Research

Exhibit 4: RMB settlement leads invoicing, especially in China-related trade



RMB invoicing shares are as of 2023; RMB settlement/payment shares are as of 2025.

Source: Boz et. al. (2025), Wind, Goldman Sachs Global Investment Research

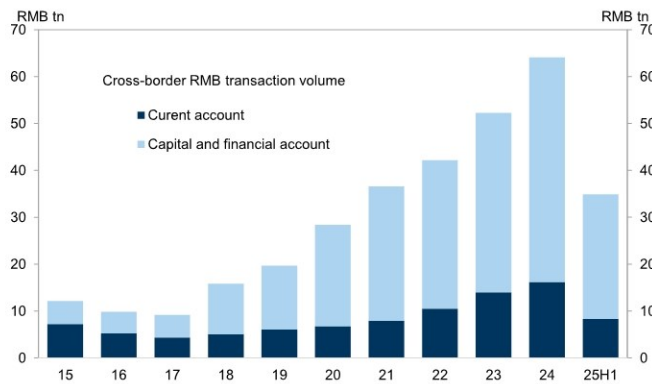
2. Financial flows dominate cross-border RMB transactions

We next look at China's cross-border RMB transactions. After falling in 2015-2017 amid RMB depreciation pressure after [the 2015 FX reform](#) and tighter management of cross-border flows, total cross-border RMB transactions rose sharply from 2017 onward, reaching RMB 64 trillion in 2024, up from RMB 9 trillion in 2017 ([Exhibit 5](#)). The composition has also shifted meaningfully. The current-account share fell sharply after 2015 and has stayed around 25% in recent years, while capital and financial account transactions now account for the other 75% of China's cross-border RMB transactions.

Bond investment is now the largest use case for cross-border RMB transactions, including Bond Connect and direct access to China Interbank Bond Market (CIBM Direct). In 2024, it accounted for 46% of cross-border RMB transactions, compared with

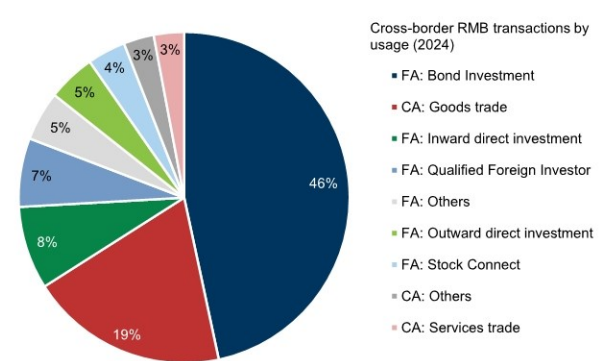
19% for goods trade and 13% for direct investment (Exhibit 6). The Qualified Foreign Investor (QFI) program accounted for another 7%, while Stock Connect accounted for 4%. This breakdown suggests that RMB cross-border use has become increasingly driven by investment flows. Therefore, the next phase of RMB internationalization will require more than trade settlement channels: foreign participants will also need more stable offshore liquidity, better risk management tools and a deeper RMB asset pool.

Exhibit 5: Financial flows have become the main driver of cross-border RMB transactions



Source: PBOC, Wind, Goldman Sachs Global Investment Research

Exhibit 6: Bond investment accounted for nearly half of cross-border RMB transactions in 2024

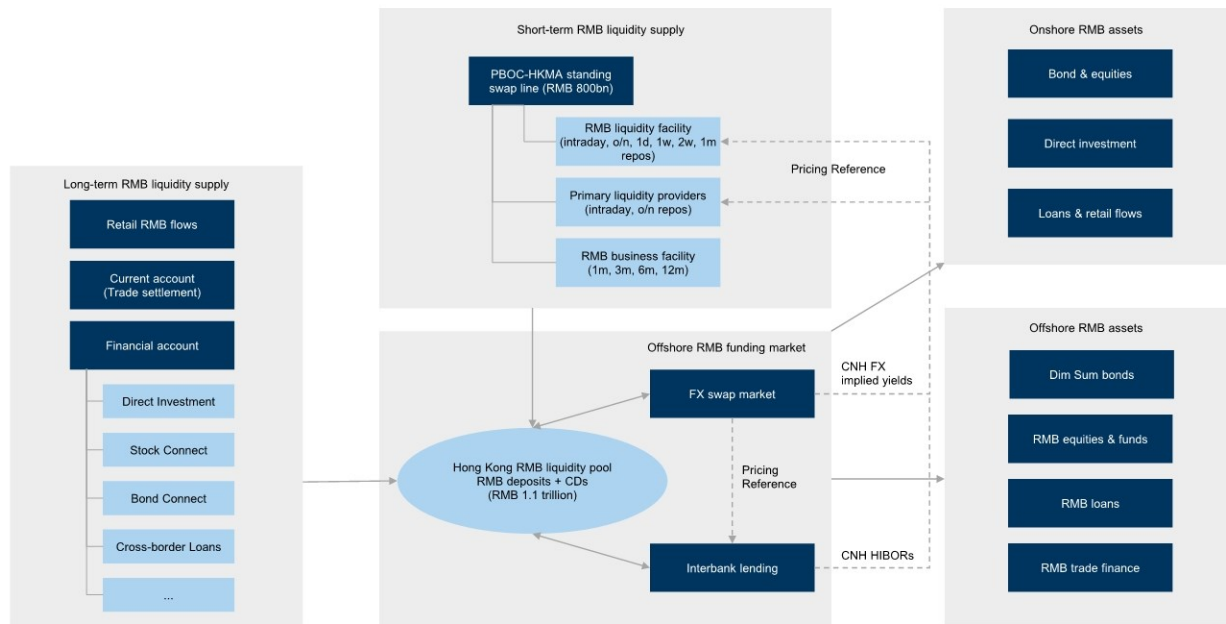


Source: Wind, Goldman Sachs Global Investment Research

What would it take for the RMB to move beyond China-linked trade settlement?

The next phase of RMB internationalization is likely to rely on gradual onshore opening and deeper offshore RMB markets. Hong Kong can function as a policy sandbox, where broader capital-account opening remains gradual and controlled, while offshore RMB funding, trading, and hedging channels continue to expand.

Exhibit 7 shows how offshore RMB liquidity circulates through Hong Kong. The left block captures medium-to-long term RMB supply from retail flows, trade, investment, and cross-border lending. The top block captures official backstop, including HKMA RMB liquidity facilities, which are ultimately supported by the PBOC-HKMA standing swap line. These channels feed into Hong Kong's RMB liquidity pool, mainly deposits and certificates of deposits, which are then deployed into offshore RMB lending, dim sum bonds, RMB derivatives and onshore assets through various connect programs.

Exhibit 7: A framework for offshore RMB liquidity in Hong Kong

Source: HKMA, Goldman Sachs Global Investment Research

The framework highlights a key point: offshore RMB market development is not only about accumulating RMB deposits or settling China-linked trade. It requires an ecosystem in which RMB can be funded, invested, hedged and redeployed. To move beyond China-linked trade settlement, the RMB therefore needs to become easier to fund, more convenient to hedge and more attractive to hold. We focus on three market foundations: offshore RMB liquidity, risk management tools and a broader RMB asset pool.

1. More stable offshore RMB liquidity

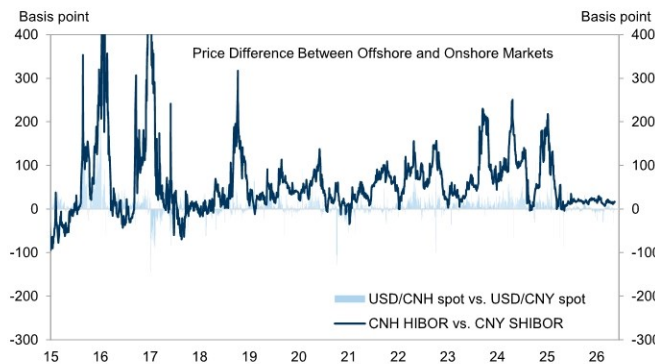
Offshore RMB liquidity has become cheaper and more stable, partly helped by recent RMB appreciation against the Dollar. Funding cost matters because it determines how easily RMB can be used for investment, lending and bond issuance offshore. In the past, investors had to factor in volatile onshore-offshore funding gaps when trading offshore RMB assets. Short-term offshore RMB funding rates in Hong Kong could be around 2pp higher than onshore RMB funding rates in 2023-24, nearly doubling the costs when onshore funding rates were around 2%.¹ The funding gaps are closely linked to RMB exchange rates. When RMB depreciation pressure rises, CNH often weakens faster than CNY against the USD, causing CNH to trade at a discount to CNY and pushing offshore RMB liquidity back onshore. In addition, PBOC FX management may also tighten offshore RMB liquidity by raising front-end CNH funding costs to discourage CNH shorts.² Together, these forces pushed up CNH funding levels and volatility in 2022-24. More recently, these frictions have eased: the spread between 3-month CNH HIBOR and

¹ We use 3 month CNH HIBOR and SHIBOR as simple proxies for offshore and onshore RMB funding costs, respectively. Both are unsecured interbank lending reference rate in HK/mainland China. Therefore the spread between them is a useful gauge of the onshore-offshore RMB funding gap.

² CNH and CNY operate as parallel currencies under onshore capital controls. While CNH is used offshore and CNY onshore, policymakers aim to keep the two close to parity. See [Bahaj and Reis \(2024\)](#) for details.

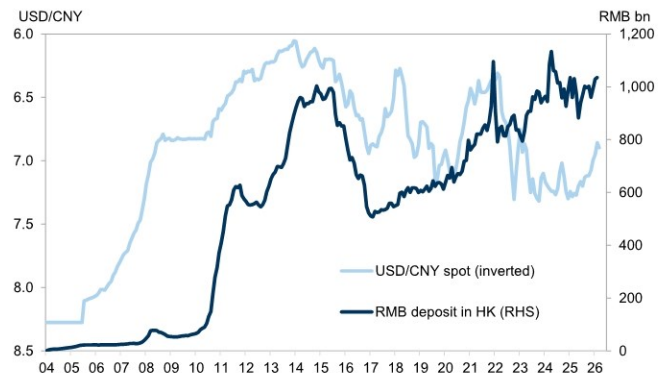
3-month SHIBOR has narrowed to around 15-25bp since April 2026, compared with 100-200bp and much higher volatility previously ([Exhibit 8](#)). Hong Kong RMB deposits also tend to rise during RMB appreciation cycles, as CNH often appreciates more than CNY ([Exhibit 9](#)).³ We expect USD/CNY to reach 6.50 in 12 months, supported by policy intention, export strength and significant FX undervaluation. Continued RMB appreciation should help keep onshore-offshore funding gaps narrow and stable.

Exhibit 8: CNH-CNY funding spreads have narrowed and stabilized since April 2025



Source: Goldman Sachs FICC and Equities, Bloomberg, CEIC

Exhibit 9: Hong Kong RMB deposits tend to rise during RMB appreciation cycles



Source: Haver Analytics, Wind

The source of offshore RMB liquidity in Hong Kong has also changed. Since the launch of RMB cross-border trade settlement between mainland China and HK in July 2009, trade settlement was an important source of RMB liquidity in HK until 2019. More recently, trade-related RMB remittances have increasingly flowed from Hong Kong back to mainland China (vs. RMB300-400bn trade-related remittances into Hong Kong in 2017-19). This suggests that Hong Kong's RMB liquidity pool has become more dependent on financial flows, such as cross-border financing, direct investment, portfolio investment and the PBOC-HKMA RMB swap line.

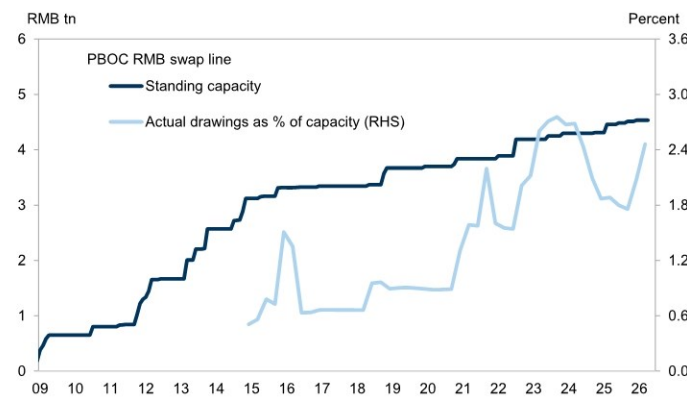
Recent policy measures continue to support cross-border RMB financial flows. In February 2025, the HKMA launched the RMB Business Facility to support RMB trade financing in Hong Kong, priced with reference to onshore RMB funding costs. In September 2025, the launch of cross-boundary bond repo business allowed foreign investors to participate in the onshore repo market and remit the RMB liquidity obtained for offshore use. In February 2026, the PBOC eased cross-border RMB interbank financing rules by moving from separate business-specific quotas to a unified net lending quota for onshore financial institutions, giving banks more flexibility to provide RMB funding offshore. Together with the offshore RMB bond repo business launched in February 2025, these measures should help make offshore RMB liquidity more stable and less costly.

Official infrastructure also matters. Offshore RMB clearing banks and PBOC swap lines provide the operational infrastructure and emergency liquidity support for offshore RMB use. Clearing banks help offshore banks and companies to settle RMB payments

³ RMB appreciation expectations are not the only driver of Hong Kong RMB deposit growth. The 2023-24 period is a useful caveat: deposits rose despite RMB depreciation pressure, partly because low CNH funding rates (vs. USD funding rates) supported faster offshore RMB loan growth. Institutional channels of long-term RMB supply also matter. Stock Connect, Bond Connect, cross-boundary repo arrangements, HKMA RMB liquidity facilities and the PBOC-HKMA swap line have all helped create more RMB deposits offshore.

locally, without routing every transaction back through mainland China. This network has expanded from the first offshore RMB clearing bank in Hong Kong in 2003 to 35 clearing banks in 33 economies by June 2025. Economies covered by offshore RMB clearing bank arrangements accounted for about 64% of China's goods trade in 2025, based on our estimates. PBOC swap lines are an official RMB liquidity backstop: a foreign central bank can obtain RMB from the PBOC and provide it to local banks when needed.⁴ Swap line capacity has expanded rapidly since 2009, but actual drawings remain relatively low (around 2–3% of total capacity; [Exhibit 10](#)). [PBOC RMB swap line partners](#) accounted for around 60% of China's goods trade in 2025. Unlike the Fed's dollar swap network, which mainly addresses dollar funding stress, PBOC swap lines have broader goals of supporting bilateral trade, investment and RMB international use. As discussed above, Hong Kong's RMB liquidity shows how swap lines can support offshore RMB funding. Perez-Saiz and Zhang (2023) [find](#) that swap lines and offshore RMB clearing banks have helped boost RMB use in cross-border payments with China, partly by easing offshore RMB liquidity constraints under China's capital controls.

Exhibit 10: PBOC RMB swap lines have expanded in coverage and capacity, while utilization remains low



Source: PBOC, Wind, Data compiled by Goldman Sachs Global Investment Research

2. Better risk management tools

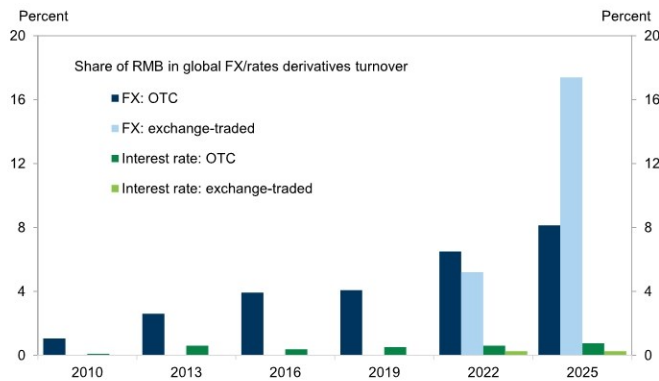
Foreign investors need tools to manage FX, rates and credit risks if they are to hold RMB assets, raising RMB funds or doing business in RMB. FX hedging manages currency moves; rates hedging manages benchmark interest-rate risk and funding costs; and credit hedging manages credit-spread risk in RMB credit exposure. RMB risk-management tools have developed unevenly across markets. The RMB's footprint in global derivatives markets is much larger in FX than in rates: according to [the BIS Triennial Survey](#), the RMB's share of global OTC FX derivatives (such as FX swaps) turnover rose from 1.0% in 2010 to 8.1% in 2025, while its share of global OTC interest-rate derivatives turnover increased only from 0.1% to 0.8% over the same period and has changed little since reaching around 0.6% in 2013 ([Exhibit 11](#)).⁵ The

⁴ A PBOC RMB swap line is a bilateral currency swap agreement between the PBOC and another central bank. When activated, the foreign central bank provides its local currency to the PBOC as collateral and receives RMB, which can then be on-lent to local financial institutions to support RMB payments. The RMB is repaid at maturity at a pre-agreed interest rate. The swap lines are usually signed for a term of three years and often renewed upon maturity.

⁵ OTC derivatives are bilateral contracts negotiated with brokers (often banks), such as FX swaps and interest rate swaps (IRS); exchange-traded derivatives are standardized contracts traded on exchanges, such as FX and

slower development of RMB rates derivatives partly reflects China's gradual interest-rate liberalization, evolving benchmark rate framework and the late relaunch of CGB futures in 2013.⁶ FX hedging is therefore the more developed part of the RMB toolkit, though CNY/CNH basis risk remains an important consideration for offshore investors (Exhibit 12).⁷ For example, in 2023–24, hedging RMB depreciation risk in the offshore market could earn around 2pp per year from the hedge, compared with around 4pp per year in the onshore market.

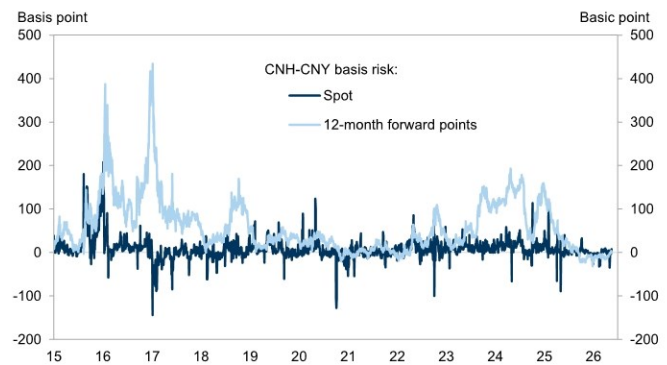
Exhibit 11: RMB's footprint in global derivatives markets is much larger in FX than in rates



Exchange-traded RMB FX and interest rate derivatives data are available only for the 2022 and 2025 survey years.

Source: BIS, Goldman Sachs Global Investment Research

Exhibit 12: The onshore-offshore spread of FX forward points remains significant, while the basis risk for FX spot is limited



We calculate offshore price minus onshore price as basis points of USD/CNY spot.

Source: Goldman Sachs FICC and Equities

Box: RMB FX hedging: swaps dominate, futures complement

RMB FX derivatives are the more developed part of the RMB hedging toolkit. The RMB's share of global OTC FX derivatives turnover rose from 1.0% in 2010 to 8.1% in 2025, far above its 0.8% share in global OTC interest-rate derivatives turnover. RMB FX derivatives are highly concentrated: around 96% is traded against the USD, and FX swaps account for around 68% of turnover.⁸ More than 90% of offshore RMB trading takes place in four centres — Hong Kong, Singapore, the UK and the US. Hong Kong remains the largest offshore RMB trading hub, accounting for more than 40% of OTC RMB FX derivatives turnover

bond futures.

⁶ Official loan rate floor and deposit rate ceiling were removed in 2013 and 2015, respectively. De jure, the interest rate system has been liberalized, though de facto this is not the case, as banks still rely heavily on LPR in their pricing model and the ceiling of deposits rates is still regulated by the "deposit rate self-regulatory mechanism". The PBOC has also moved gradually toward a more rate-based framework, with DR007 becoming an important operational reference around 2019, the 7-day reverse repo rate later clarified as the primary policy rate, and temporary overnight OMO operations more recently creating a de facto narrower corridor around the 7-day reverse repo rate. The central bank appears to have shifted further toward overnight repo rates as operational reference since early 2025.

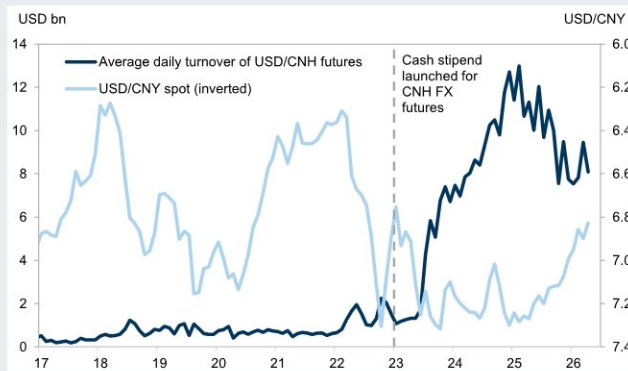
⁷ USD/CNY spot and USD/CNH spot usually trade close to each other, with an average onshore-offshore basis of around 10bp in 2022–25. But the basis in FX forward points can be much larger and more volatile, averaging around 60bp in 2022–25 and at times staying above 100bp for a while. When this happens, onshore hedging looks more attractive on pricing, but can still involve more operational costs, such as matching hedges to underlying onshore portfolio exposure and meeting onshore documentation and trading-processing requirements.

⁸ An FX swap works like collateralized currency funding: an investor provides one currency today, receives

outside mainland China. Robbert, Sushko, and Westermann (2026) find that this geographical concentration increasingly reflects financial links with China, including cross-border banking exposure to China and Qualified Foreign Investors (QFI) access.

Exchange-traded RMB FX products (such as FX futures) remain a much smaller market compared with OTC derivatives (such as FX swaps). Globally, exchange-traded FX derivatives turnover has been only around 3% of OTC FX derivatives turnover over the past decade. For RMB, exchange-traded FX derivatives reached 7% of OTC turnover in 2025, up from 2% in 2022. Hong Kong's USD/CNH futures market illustrates the point. Turnover remained limited for much of the period after launch in 2012, but rose notably after market-maker incentives improved liquidity provision from 2023 (Exhibit 13).

Exhibit 13: Market-maker incentives helped lift USD/CNH futures turnover in Hong Kong



Source: CEIC, Haver Analytics

RMB FX futures are unlikely to replace OTC FX swaps, but they can complement the market by broadening access and improving price discovery. OTC products remain better suited for large, customized hedging needs; futures offer standardized contracts, transparent pricing, centralized clearing and settlement, and easier operational access for some investors and corporates. Recent policy discussion on developing onshore RMB FX futures points in the same direction.

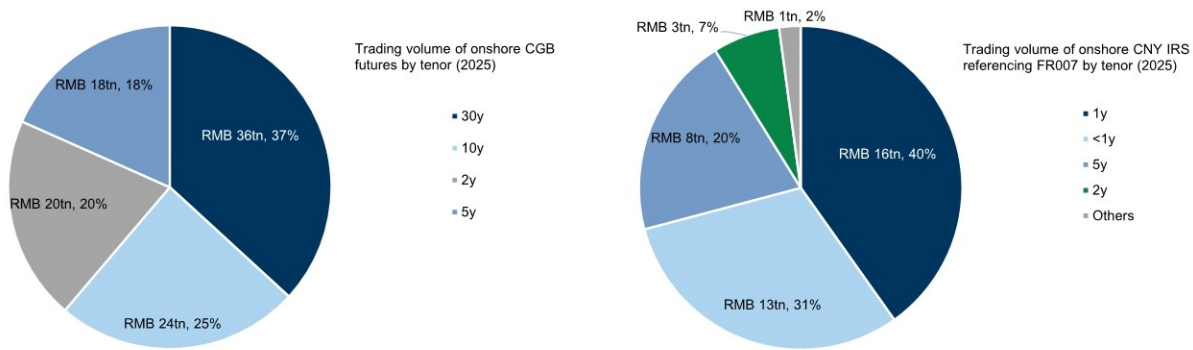
The bigger gap is long-duration rates hedging for foreign investors. The main tools are interest rate swaps (IRS), which are OTC products, and Chinese government bond (CGB) futures, which are exchange-traded products. IRS is available both onshore and offshore, while CGB futures currently trade only onshore. Note that offshore IRS refers to non-deliverable interest rate swaps (NDIRS), which are settled in USD on a net cash basis without exchanging RMB. IRS and CGB futures hedge different risks: IRS is linked to funding rate expectations, while CGB futures are more directly linked to government bond yield curve. Onshore IRS liquidity is concentrated in the front-end and 5y, while liquidity beyond 5y is limited. By contrast, onshore CGB futures are mostly traded at longer tenors, with 30y and 10y contracts accounting for 37% and 25% of turnover, respectively (Exhibit 14). Hedge effectiveness points in the same direction. Exhibit 15 shows that CGB futures have tended to provide a closer hedge for long-end CGBs than IRS or NDIRS over the past few years.⁹ IRS can still track long-end CGB yields well

another currency, and reverses the exchange at a pre-agreed rate in the future.

⁹ A more rigorous hedge-effectiveness test would regress CGB price changes on changes in the hedging instrument, such as CGB future prices and dollar value of IRS. A significant, high and stable beta would suggest

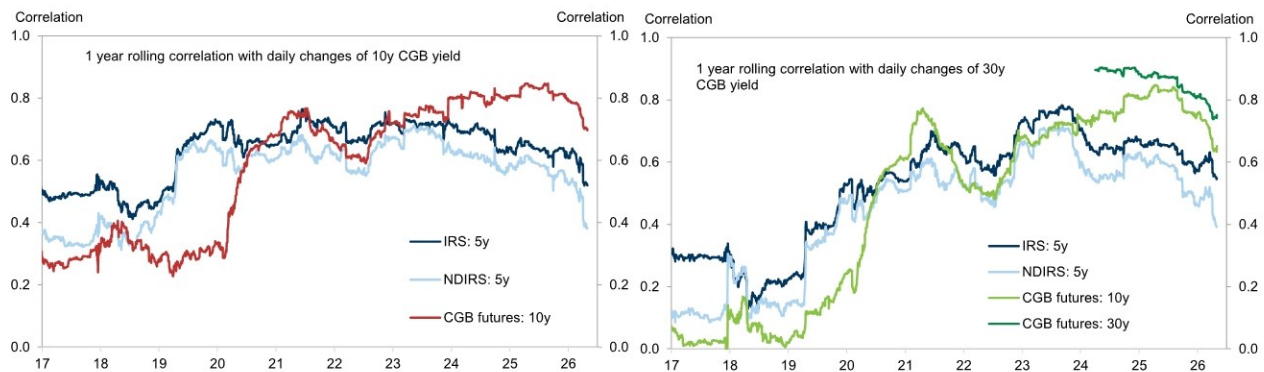
because both reflect domestic funding conditions and/or monetary policy expectations.

Exhibit 14: CGB futures trading is concentrated in 10y and 30y tenors, while IRS trading is mostly at the front end



Source: CEIC, Goldman Sachs Global Investment Research

Exhibit 15: Longer-dated CGB futures provide a better hedge for long-end CGBs



Source: Goldman Sachs FICC and Equities, CEIC, Goldman Sachs Global Investment Research

Policy has started to address these gaps. Since 2020, foreign investors in China's interbank bond market have been allowed to use onshore FX derivatives to hedge FX risks from bond investment, and the extension of CFETS FX trading hours to 3am Beijing time has made onshore trading more accessible across global time zones. Swap Connect, launched in 2023, gave offshore investors access to onshore IRS, which narrowed the spread between IRS and NDIRS rates. In April 2026, qualified foreign investors were allowed to trade onshore CGB futures for hedging purposes. Hong Kong is also preparing an offshore CGB futures market. These steps will provide a broader toolkit that can better support long-duration hedging for foreign investors. Credit risk hedging is a further gap. Market feedback suggests that defaults in onshore credit bonds and insufficient risk-mitigation tools remain concerns for overseas investors. For foreign investors to expand beyond CGBs and policy bank bonds, they would need improved credit rating systems and more developed credit-risk mitigation tools.

3. Broader RMB asset pool

A broader RMB asset pool requires both smoother access to onshore assets and deeper offshore RMB-denominated assets. The issue is not simply the size of China's domestic markets. China's onshore equity and bond markets are already large and continue to

the hedging instrument consistently covers a large share of bond price movements. Simple correlations are useful as a first look at whether IRS/CGB futures are directionally aligned with changes of CGB yields.

expand, with A-share market capitalization at around RMB120trillion (or US\$17tn), and bond market custody balance close to RMB200trillion (or US\$28tn). Yet foreign participation has not risen mechanically. Foreign investors have reduced RMB bond holdings while increasing RMB equity holdings in recent years, and their overall share of domestic assets has declined ([Exhibit 16](#)). This partly reflects unfavorable interest rate differentials between China and major developed markets. Further enhancement of Connect programs would help encourage investment in RMB assets. For example, Swap Connect program currently only supports northbound trading; [adding southbound trading](#) could help onshore investors manage offshore RMB rates exposure, encourage more Southbound allocation to offshore RMB assets and add depth to Hong Kong's offshore RMB liquidity pool. Expanding Connect programs to a broader set of assets, such as real-estate investment trusts (REITs), private funds and commodity futures, could also broaden the investable RMB universe, and thus potentially raise RMB international use.

Exhibit 16: Foreign investors have increased their holdings of RMB equities but reduced holdings of RMB bonds over the past few months



Source: PBOC, Wind, Data compiled by Goldman Sachs Global Investment Research

Offshore RMB-denominated assets also need to deepen. The offshore market still lacks a liquid CGB yield curve. Dim sum bonds account for only a small share of the international debt securities market, and the outstanding amount was only around RMB1.4trillion as of April 2026. Further growth would require a broader issuer base, better secondary-market liquidity and longer-dated issuance. There has been some progress: some China tech companies (such as JD and Tencent) have [issued](#) dim sum bonds over the past two years. Foreign financial institutions have also [become more active](#), issuing longer-dated bonds. More offshore RMB government bond issuance in Hong Kong could [help establish](#) a more liquid yield curve. Hong Kong's plan to [develop a commodity trading ecosystem](#) could also help over time, especially if more RMB-denominated products are introduced.

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Disclosure Appendix

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